

Executive Summary

Customer churn is a major challenge for subscription-based businesses, as losing existing customers can significantly impact long-term revenue and profitability. This project aims to identify customers who are at risk of churning and uncover the key factors influencing their decision to leave. By leveraging historical customer information, transaction records, and user activity data, predictive analytics can help businesses take proactive actions to improve customer retention.

To achieve this objective, a **LightGBM** classification model was developed to predict the likelihood of customer churn. The model analyzes customer demographics, subscription history, payment behavior, pricing changes, membership duration, and listening activity to estimate each customer's probability of leaving the service. In addition to generating accurate churn predictions, Explainable AI (SHAP) was used to interpret the model's decisions and identify the most influential factors driving customer churn.

The primary business objective of this analysis is to **reduce customer churn by identifying at-risk customers and understanding the key drivers behind churn**. The insights presented in this report translate the model's predictions into actionable business recommendations that can support customer retention strategies, improve marketing effectiveness, optimize subscription plans, and strengthen long-term customer loyalty.

This report focuses exclusively on the **business analysis** of the project, presenting visual insights, model explanations, and strategic recommendations. Detailed information regarding data preprocessing, feature engineering, and model development is documented separately in the accompanying technical notebooks.

KKBOX Customer Churn Prediction

Business Intelligence Dashboard

Note: All business insights, visualizations, and recommendations in this report are based on the **unseen test dataset**, ensuring that the analysis reflects the model's performance on data that was never used during training.

How healthy is the customer base?

Executive KPI Dashboard

Note: The Revenue at Risk metric is an illustrative estimate based on an assumed average monthly subscription value of \$149 per customer.

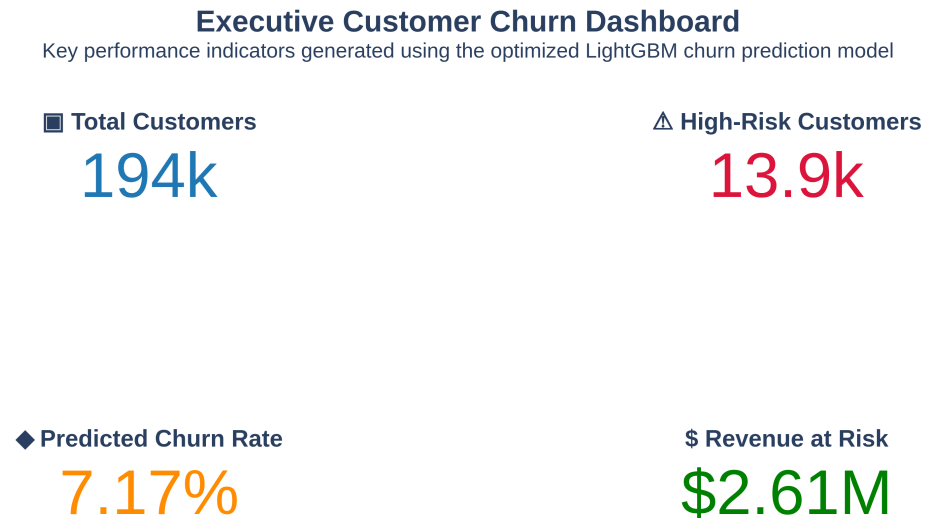


Figure 1: Executive_KPI_Dashboard (1).png

Business Insight

- The model analyzed 194K customers and identified 13.9K customers as high-risk of churn.
- 7.17% of customers were classified as high-risk by the churn prediction model and should be prioritized for retention initiatives. These customers represent the highest-priority segment for proactive retention efforts.
- Based on an assumed average monthly subscription value of \$149 per customer, the potential revenue at risk is estimated at approximately \$2.61M. This figure is illustrative and demonstrates how churn predictions can be translated into business impact.

Business Recommendations

- Prioritize personalized retention campaigns for high-risk customers before their subscriptions expire.

- Offer targeted incentives (discounts, premium trials, or loyalty rewards) only to customers with the highest predicted churn probability. Continuously monitor model predictions to measure retention performance and update intervention strategies.
- Focus marketing resources on customers most likely to churn to maximize return on retention spending.

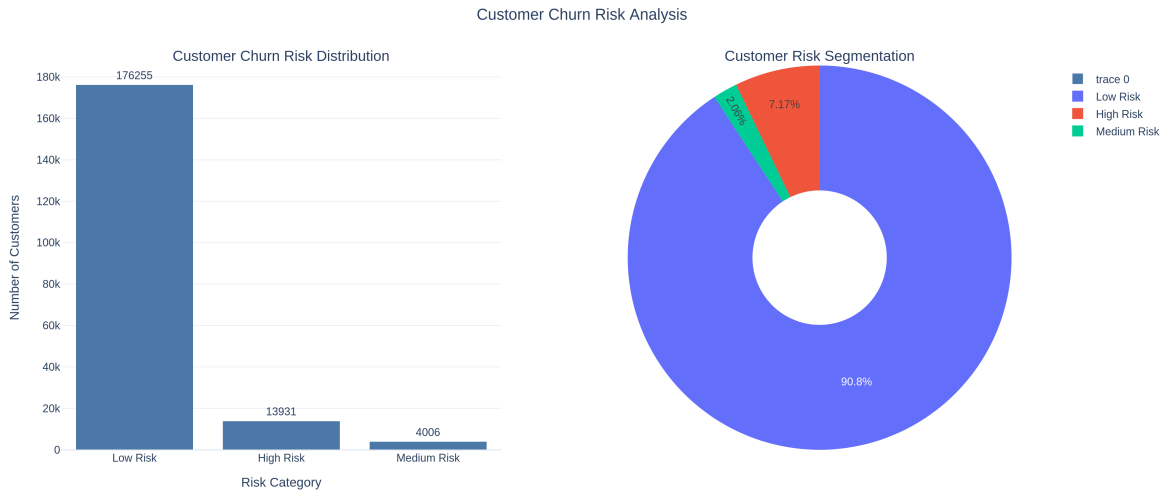


Figure 2: chart_02.png

Business Insight

- Most customers (90.8%) are classified as Low Risk, indicating a generally stable customer base.
- Only 7.17% of customers fall into the High-Risk segment, enabling retention efforts to focus on a relatively small, high-priority group.
- The Medium-Risk segment (2.06%) represents customers who could potentially transition to high risk without timely engagement.

Business Recommendations

- Prioritize retention campaigns for High-Risk customers while monitoring Medium-Risk customers for early warning signs.
- Maintain customer satisfaction programs for the Low-Risk segment to preserve long-term loyalty.
- Review risk segment distributions regularly to measure the effectiveness of retention strategies and detect shifts in customer behavior.

Customer Behavior Analysis

Which customer behaviors indicate churn?

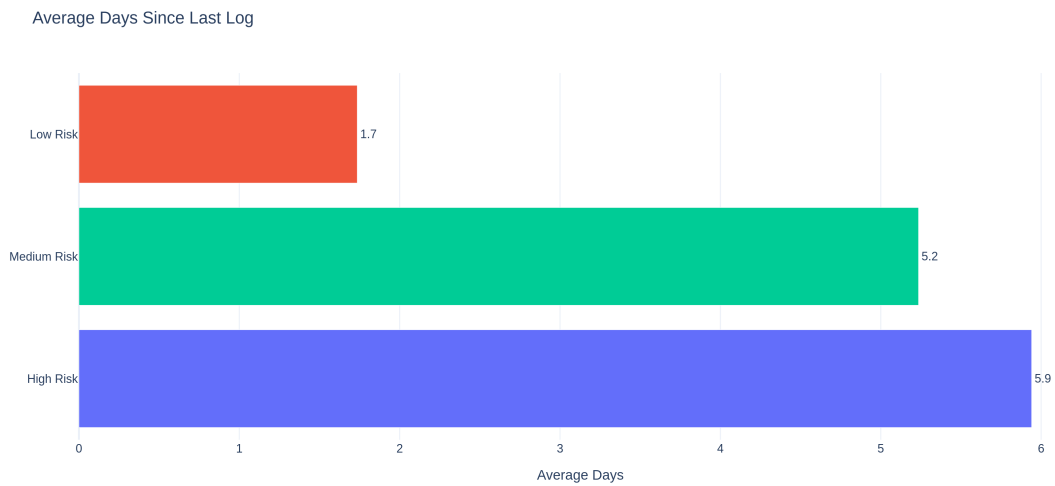


Figure 3: chart_03.png

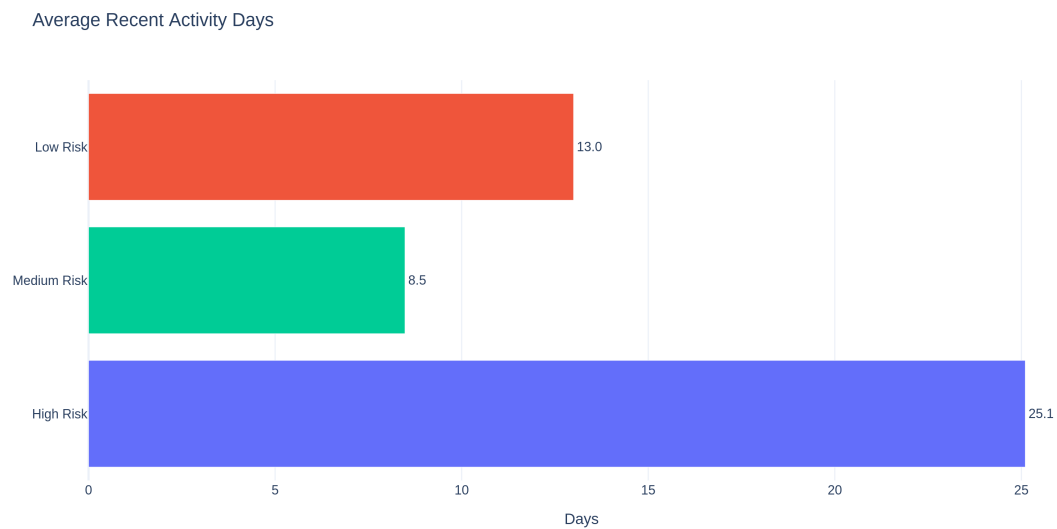


Figure 4: chart_04.png

Business Insight

- High-risk customers exhibit significantly lower engagement. On average, they remain inactive for nearly twice as many days as low-risk customers and take longer to return to the platform. Declining engagement is a strong early indicator of potential churn.

Business Recommendation

- Monitor customer inactivity using Recent Activity Days and Days Since Last Log as early warning indicators.
- Trigger automated re-engagement campaigns (personalized emails, push notifications, or content recommendations) when customers become inactive.
- Prioritize retention efforts for customers showing sustained declines in platform engagement.

Subscription Analysis

Which subscription patterns increase churn risk?

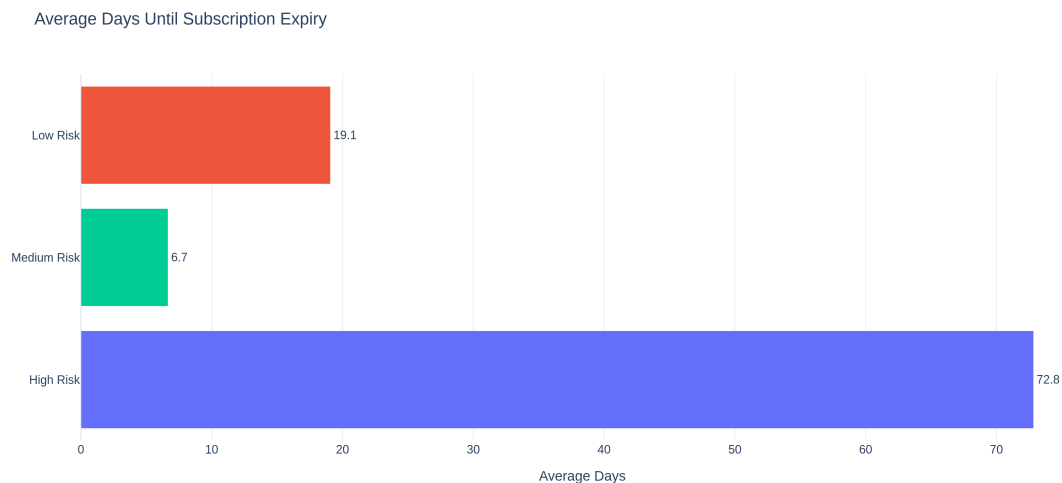


Figure 5: chart_05.png

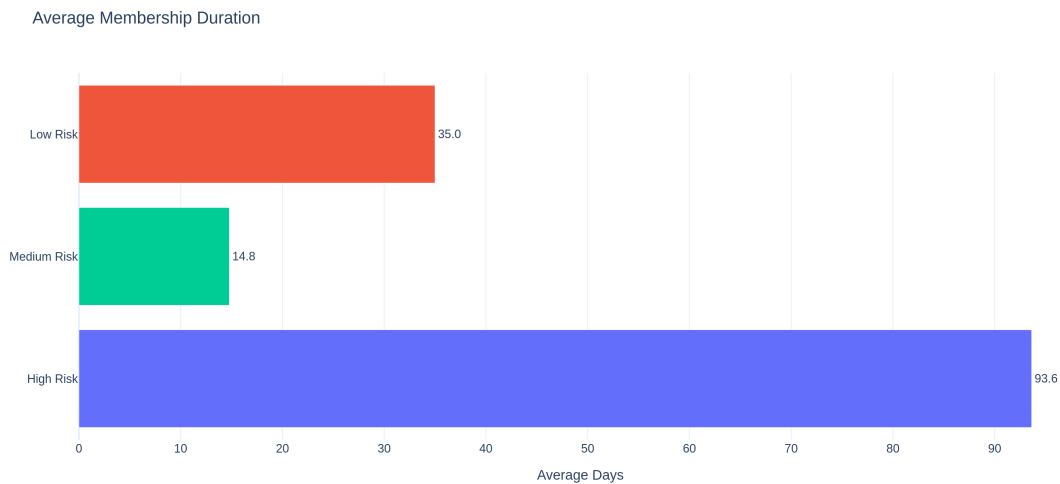


Figure 6: chart_06.png

Business Insight

- High-risk customers have maintained subscriptions much longer than other groups, indicating that churn is often driven by long-term disengagement rather than by new customers leaving immediately.
- High-risk customers still have, on average, over two months remaining before subscription expiry. This suggests many customers disengage long before their subscription ends, providing a valuable window for proactive retention campaigns.

Business Recommendation

- Identify long-term subscribers who have become disengaged and offer loyalty incentives or exclusive benefits before renewal decisions are made.
- Monitor customers with substantial remaining subscription time but declining activity, as they may require proactive intervention despite not being close to expiration.
- Segment retention campaigns based on subscription lifecycle, ensuring experienced subscribers receive different offers than newer customers.

Customer Segmentation

Which types of customers are more likely to churn?

Age Risk Analysis

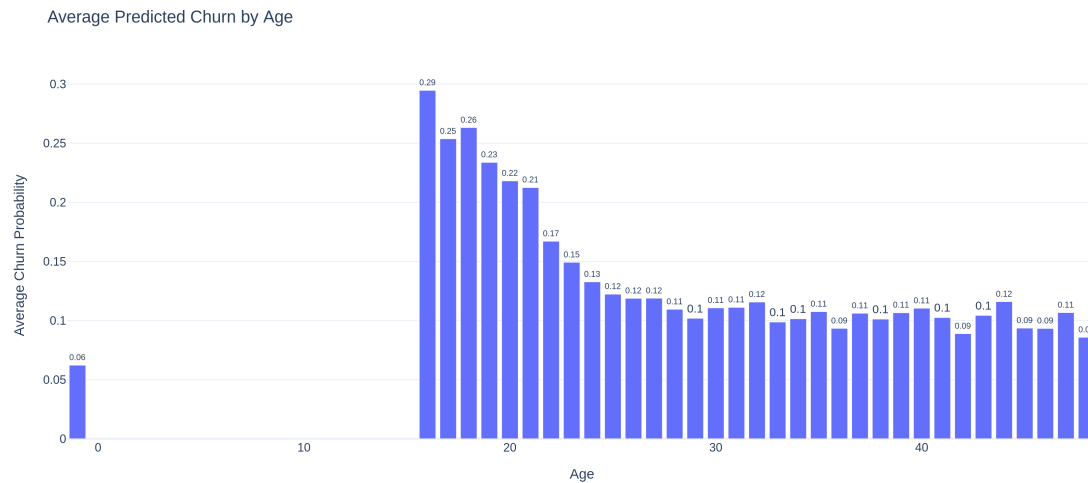


Figure 7: chart_07.png

Business Insight

- Customers aged 16–22 have the highest predicted churn probability, with risk gradually decreasing as age increases.
- After approximately 25 years old, churn becomes relatively stable with only small differences between age groups.
- This indicates that younger customers are substantially more volatile, while older customers exhibit more consistent subscription behavior. The isolated value for age -1 has a very small probability and it represents missing or unknown age rather than a real customer segment, so it should not drive business decisions.

Business Recommendation

- Develop a youth-focused retention strategy instead of applying identical campaigns to every customer.
- Introduce personalized onboarding, early engagement campaigns, and student or young-user promotions during the first months after registration.

- Allocate retention budget preferentially toward younger customers because they present the greatest opportunity for reducing future churn.

Acquisition Channel Analysis

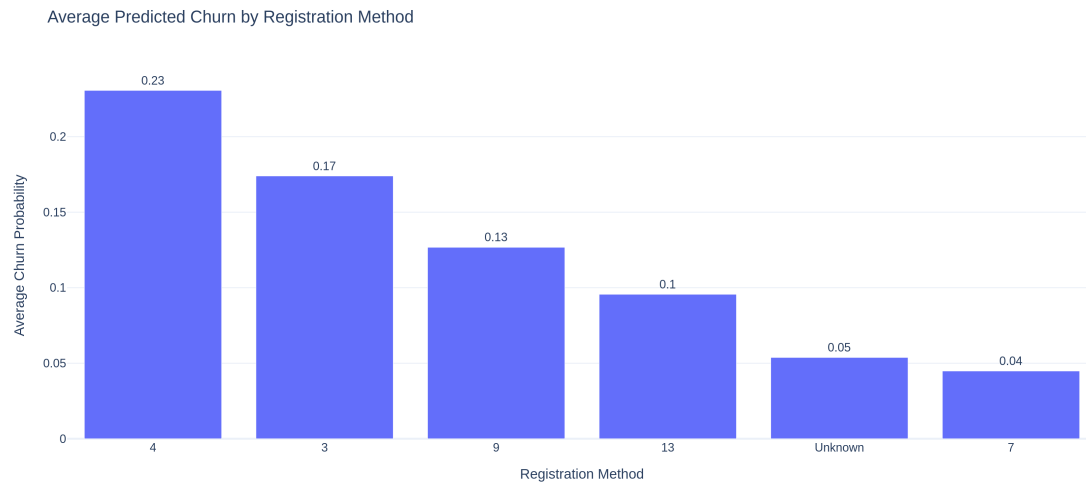


Figure 8: chart_08.png

Business Insight

- Registration methods differ noticeably in predicted churn. Method 4 produces the highest average churn probability, followed by methods 3 and 9.
- Method 7 and customers with unknown registration methods exhibit substantially lower churn.
- This suggests that customer acquisition quality varies across registration channels, meaning some channels attract less loyal users than others.

Business Recommendation

- Evaluate the marketing campaigns associated with high-risk registration channels rather than simply increasing acquisition volume.
- Compare conversion quality, onboarding experience, and customer lifetime value across registration methods.
- Optimize or discontinue acquisition campaigns that consistently attract customers with poor long-term retention.

Payment Behavior Analysis

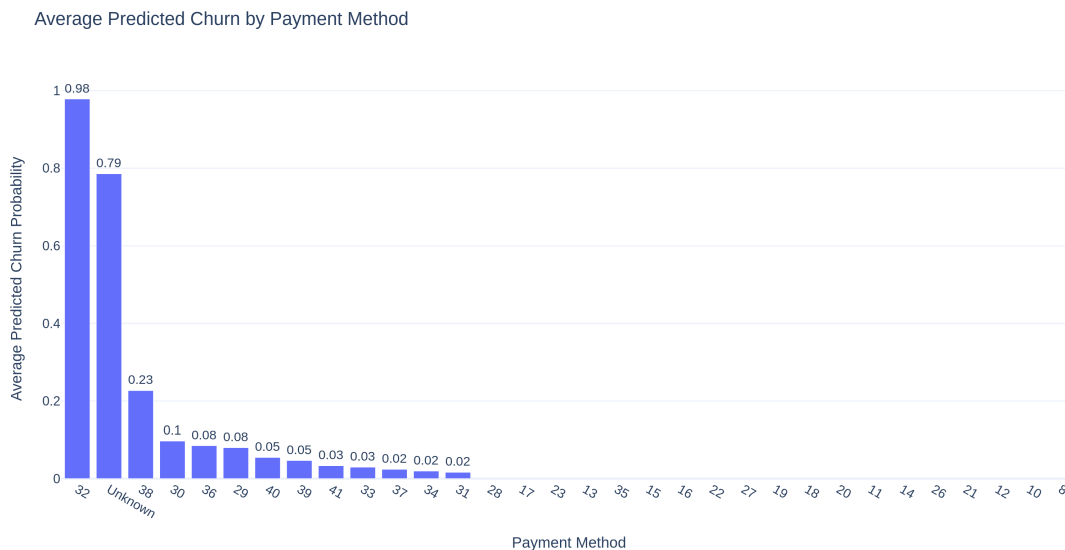


Figure 9: chart_09.png

Business Insight

- Payment method 32 exhibits an exceptionally high predicted churn probability (~98%), while customers with unknown payment methods also show elevated churn.
- Most remaining payment methods demonstrate relatively low churn probabilities, indicating that churn risk is concentrated within only a few payment categories.
- The results suggest that payment behavior is an important behavioral indicator rather than merely a billing attribute. The model identifies payment method as an early signal of future churn, implying that differences in payment workflows or customer preferences may influence retention.

Business Recommendation

- Investigate why customers using payment method 32 consistently experience extremely high churn.
- Review whether these customers encounter payment failures, short-term subscription plans, manual renewals, or poorer user experiences.
- Implement targeted retention campaigns for customers using high-risk payment methods before their subscriptions expire.

- Monitor payment-method performance continuously to detect newly emerging high-risk payment behaviors.

Top 20 Most Important Features (LightGBM)

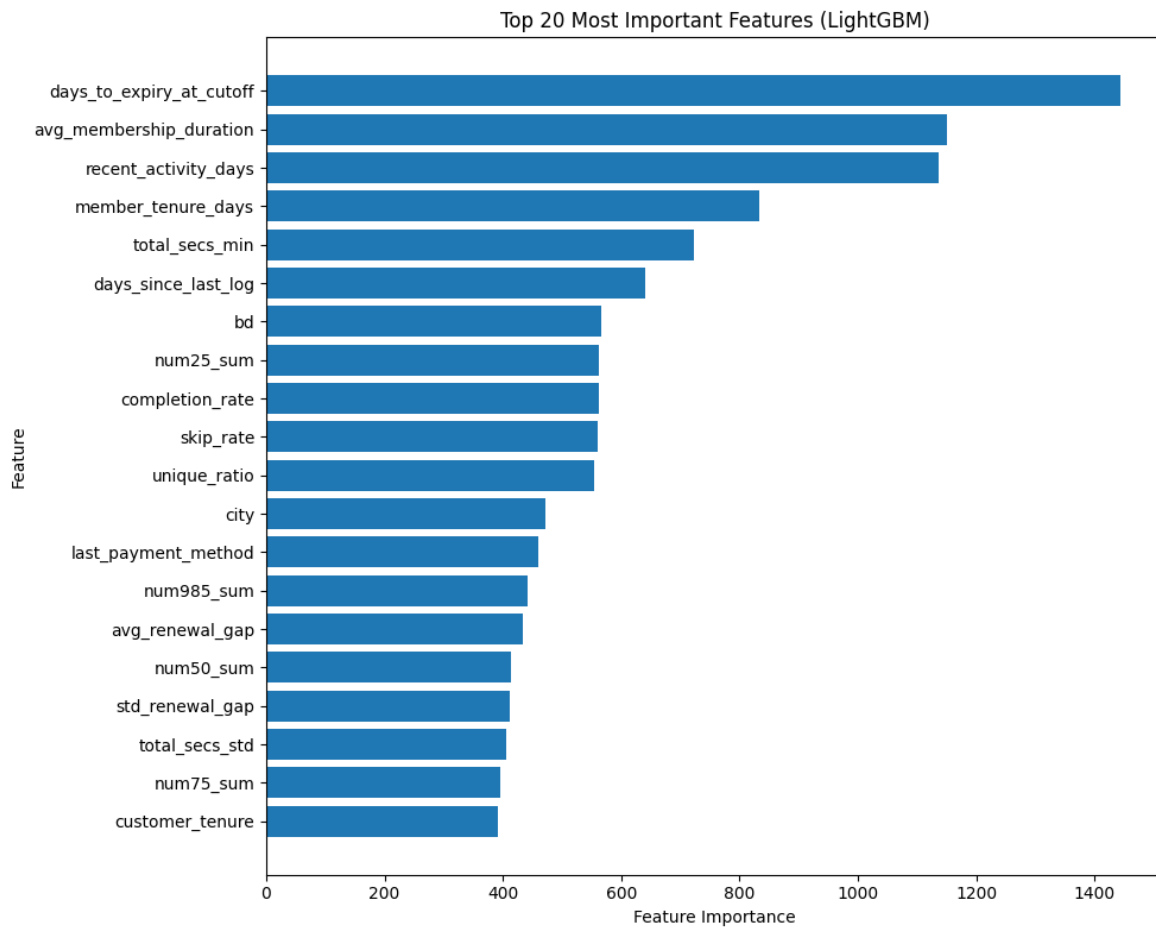


Figure 10: chart_11.png

What features drive churn across all customers?

Top 13 Global Drivers of Customer Churn

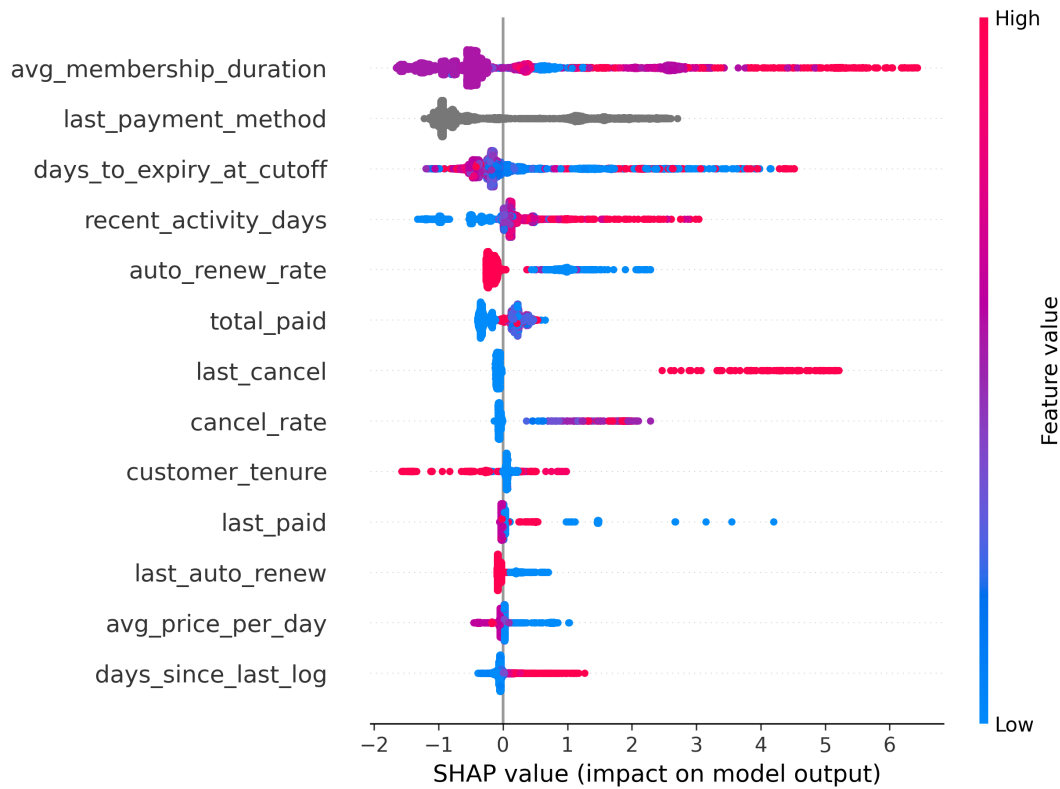


Figure 11: chart_12 (2).png

Business Insight

- The SHAP analysis reveals that the model primarily learns from customer behavior rather than demographic attributes. Instead of relying on age or registration channel, the model combines subscription history, payment behavior, renewal patterns, and cancellation signals to estimate churn risk. This demonstrates that customer engagement throughout the subscription lifecycle is far more informative than static profile information.

Business Recommendation

- Organizations should build churn monitoring systems around behavioral signals rather than demographic segmentation alone. Combining explainable AI with predictive models enables customer success teams to understand the primary drivers behind each prediction, prioritize high-risk customers more effectively, and design targeted retention strategies based on the underlying causes of churn instead of applying generic campaigns.

Why was this individual customer predicted to churn?

Explainable AI (SHAP)

To ensure model predictions are transparent and actionable, SHAP (SHapley Additive exPlanations) was used to identify the features that most strongly influence churn predictions at both the global and individual customer levels.

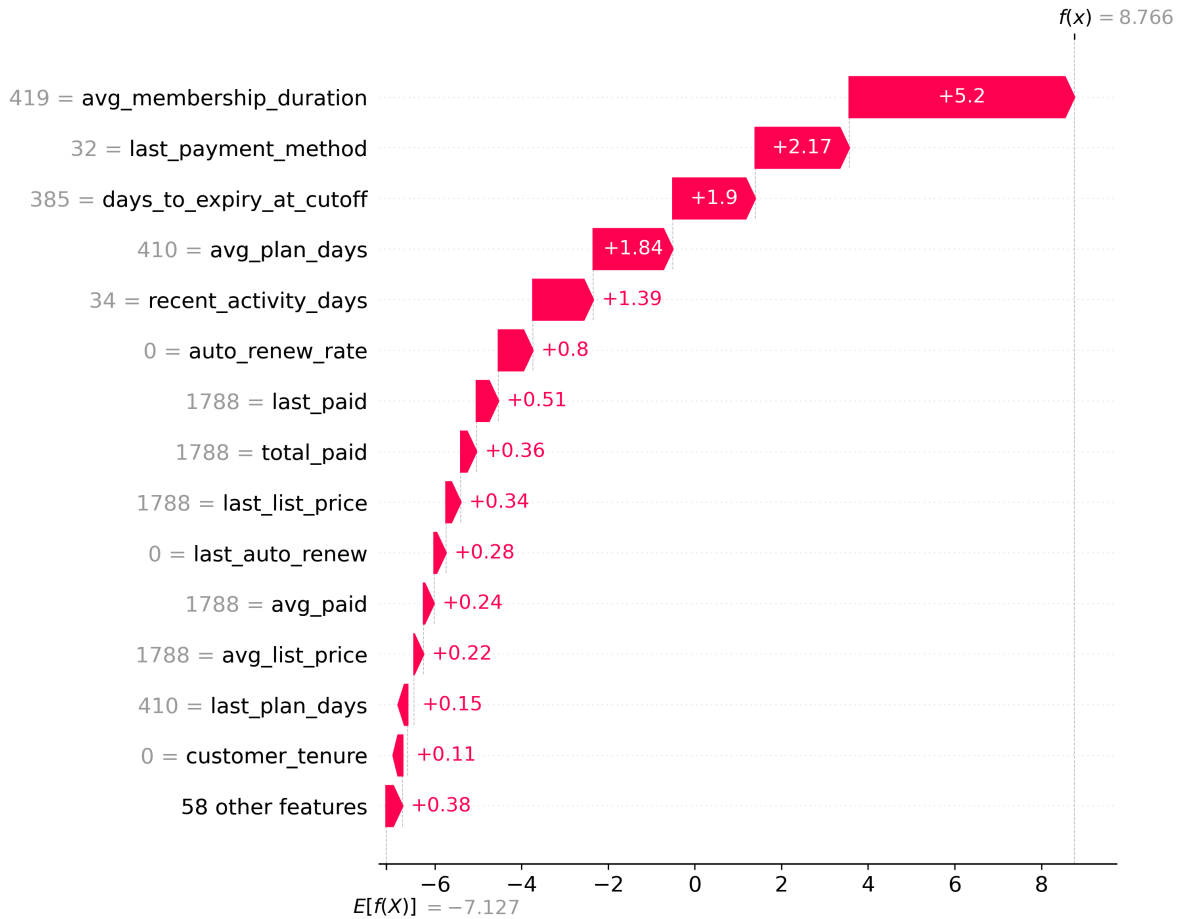


Figure 12: chart_13_clean.png

Business Insight

- The individual SHAP explanation shows that churn predictions are driven by the combined influence of multiple behavioral signals rather than any single feature. Although the customer still has an active subscription and auto-renew provides a small protective effect, the model identifies a pattern of renewal behavior, subscription history, and declining engagement that outweighs these positive factors, resulting in an extremely high churn risk.

Business Recommendation

- Instead of using one-size-fits-all retention campaigns, organizations should use explainable AI to understand the specific factors driving churn for each customer. Personalized inter-

ventions can then focus on the customer's underlying risk profile—for example, addressing declining engagement, reviewing renewal behavior, or offering loyalty incentives—rather than applying the same retention strategy to every at-risk subscriber.

Final Business Conclusion

The churn prediction model successfully identifies a relatively small subset of customers (7.17%) that represents the highest retention priority. The analysis demonstrates that churn is primarily driven by behavioral factors—including engagement, renewal patterns, payment behavior, and subscription history—rather than demographic characteristics. By focusing retention resources on these high-risk customers, the business can intervene earlier, improve customer lifetime value, and reduce potential revenue loss while using marketing budgets more efficiently.